

BUTLER COUNTY REGIONAL, OH, USAWMO: **725217**Lat: **39.364N**Lon: **84.525W**Elev: **193**StdP: **99.03**Time Zone: **-5.00 (NAE)**Period: **99-19**WBAN: **53855****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.1	-12.4	-20.3	0.6	-14.5	-17.5	0.8	-11.1	9.7	0.0	8.5	-0.6	1.5	260	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	33.5	23.7	32.3	23.3	31.1	22.8	25.4	31.3	24.7	30.0	24.0	28.8	3.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.0	28.3	23.0	18.1	27.3	22.5	17.6	26.8	78.8	31.4	75.6	30.2	72.8	28.8	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	8.2	7.2	5.9	-20.7	35.1	4.8	2.0	-24.2	36.5	-27.0	37.7	-29.7	38.9	-33.2	40.3	
(5)				-21.0	27.0	4.7	0.8	-24.4	27.6	-27.1	28.1	-29.8	28.5	-33.2	29.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	-1.0	0.7	6.2	12.7	18.0	22.2	24.1	23.4	19.5	12.8	6.5	1.5
	DBStd	10.17	6.75	6.50	6.03	5.09	4.46	3.08	2.88	2.75	4.00	4.95	5.33	5.79
	HDD10.0	1223	346	268	149	31	2	0	0	0	0	26	132	269
	HDD18.3	2813	599	495	376	179	64	6	1	1	33	183	356	521
	CDD10.0	2057	4	7	33	111	249	367	437	416	286	114	27	7
	CDD18.3	605	0	0	2	9	53	123	179	159	69	12	0	0
	CDH23.3	5963	0	1	22	125	523	1153	1754	1514	728	138	5	0
	CDH26.7	2174	0	0	3	18	144	403	709	595	271	32	1	0
Wind	WSAvg	2.3	3.0	2.9	2.9	2.9	2.1	1.8	1.6	1.5	1.6	2.0	2.3	2.6
Precipitation	PrecAvg	1056	76	62	89	108	122	110	98	77	78	76	74	88
	PrecMax	1514	155	137	239	275	265	232	224	144	177	153	173	218
	PrecMin	751	30	20	28	36	22	35	29	24	16	26	23	27
	PrecStd	176	31	32	41	56	62	43	44	31	47	38	36	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.4	21.3	26.2	28.6	31.5	33.7	35.9	35.2	34.1	30.1	23.7	18.8
		MCWB	15.0	15.3	16.4	17.9	22.0	23.6	25.7	23.8	22.1	21.8	16.1	15.0
	2%	DB	15.5	17.2	22.5	26.9	29.7	32.1	33.5	33.0	32.1	26.8	21.0	15.8
		MCWB	12.7	12.3	14.9	17.4	20.8	23.3	24.4	23.4	22.2	19.1	14.7	12.8
	5%	DB	12.5	14.0	19.2	24.5	28.3	30.9	32.1	31.7	29.9	24.3	18.5	13.0
		MCWB	10.3	10.5	13.1	16.0	20.3	22.7	24.0	23.0	21.5	17.7	13.9	10.5
	10%	DB	9.6	11.1	16.7	22.4	26.7	29.1	30.7	30.1	27.6	22.2	16.1	10.8
		MCWB	6.7	7.6	12.0	15.3	19.6	21.7	23.3	22.3	20.2	16.7	12.0	8.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	16.4	17.6	20.3	23.5	25.5	26.9	25.8	24.7	22.7	18.0	16.1
		MCDB	17.7	20.3	24.6	26.2	28.6	31.2	32.7	31.6	30.6	28.2	21.4	18.1
	2%	WB	13.4	13.9	15.9	18.8	22.3	24.5	25.8	25.1	23.7	20.8	16.2	13.2
		MCDB	15.3	16.0	20.7	23.9	27.9	30.0	31.7	30.7	29.1	24.2	19.6	14.9
	5%	WB	10.7	11.0	14.4	17.4	21.4	23.7	25.0	24.3	22.7	19.2	14.3	11.0
		MCDB	12.3	13.7	18.0	22.0	26.7	28.9	30.5	29.3	27.5	23.0	17.7	13.0
	10%	WB	7.0	7.5	12.4	16.1	20.4	22.9	24.2	23.5	21.7	17.5	12.2	8.2
		MCDB	9.1	10.6	16.3	21.0	25.0	27.6	28.9	27.9	25.6	20.6	15.4	10.1
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.5	11.4	12.9	12.0	11.7	11.4	12.2	13.4	13.1	12.1	9.1
		MCDBR	11.6	14.3	15.6	16.0	14.1	13.7	12.6	14.2	15.9	16.0	15.6	11.8
		MCWBR	9.7	10.9	9.1	8.5	6.8	6.2	5.1	5.8	7.0	8.8	10.2	9.8
	5% WB	MCDBR	10.8	13.0	12.9	13.2	11.8	11.2	11.1	11.3	12.4	12.9	12.6	10.8
		MCWBR	9.7	10.9	8.3	8.0	6.3	5.7	5.1	5.2	6.0	7.5	9.4	9.7
Clear-Sky Solar Irradiance	taub	0.307	0.328	0.355	0.398	0.441	0.454	0.459	0.454	0.402	0.358	0.329	0.306	
	taud	2.463	2.370	2.364	2.256	2.176	2.189	2.181	2.178	2.300	2.424	2.463	2.478	
	Ebn at Noon	866	890	900	877	843	830	822	815	838	845	829	839	
	Edh at Noon	79	100	112	133	147	145	145	142	118	94	79	73	
All-Sky Solar Radiation	RadAvg	1.77	2.57	3.61	4.87	5.49	6.10	5.91	5.47	4.49	3.22	2.18	1.50	
	RadStd	0.15	0.31	0.32	0.39	0.47	0.37	0.36	0.28	0.44	0.31	0.25	0.15	